



Introducing water-redispersible powderable acrylic adhesives using Persian gum

Ali Ahmadi-Dehnoei, Somayeh Ghasemirad *

Polymer Engineering Department, Faculty of Chemical Engineering, Tarbiat Modares University, P.O. Box 14115-114, Tehran, Iran

ARTICLE INFO

Keywords:

Persian gum
Powderable acrylic adhesive
Water-redispersible acrylic polymer
Ternary interaction parameter
Swelling
Adhesion strength

ABSTRACT

Persian gum (PG), as a recently-highlighted carbohydrate natural polymer, was introduced both as a powder adhesive in its pristine form, and as a water-redispersing agent for water-incompatible acrylic adhesives. Blends of PG and poly(methyl methacrylate-co-butyl acrylate) containing 30 wt. % methyl methacrylate (MBC) were molecularly incompatible at PG contents below 50 wt. % due to the large difference in their Hildebrand solubility parameters (calculated to be 37.49 and 19.71 J^{1/2}/cm^{3/2} for PG and MBC, respectively). However, further increase in the PG content evolved the system toward miscibility through intensified intermolecular hydrogen bonds. Addition of PG to MBC at compositions of more than 50 wt. % PG also provided it with powderability. Swelling of slabs of the blends in water followed a pseudo-Fickian mechanism and the equilibrium swelling ratio of the blends was a reciprocal power function of ternary interaction parameter ($\chi_{1(23)}$). Furthermore, increase in the PG content from 0 to 90 wt. % improved the adhesion strength of the blend adhesives. Ultimately, the blend containing 90 wt. % PG presented 52 times and 34 times enhancement in the adhesion strength to fabric and wood substrates compared to the pristine acrylic adhesive, respectively.

1. Introduction

Polymeric adhesives that can bond two or more parts together have been widely used in many industries, such as wood, automotive, and medicine (Liu et al., 2020). Most commercial adhesives are based on non-renewable and depleting petrochemical resources (Chen et al., 2017; Patel, 2015; Zhang et al., 2019). Moreover, several adhesives also contain residual toxic chemicals and volatile organic compounds (VOCs) (Chen et al., 2017; Kim, 2009; Patel, 2015; Zhang et al., 2019). Due to their environmental impact or hazard to living organisms' health, environmental protection organizations aim to reduce the usage of such materials (Patel, 2015; Pradyawong et al., 2018). These two problems seem to be overcome simultaneously by replacing the synthetic materials with natural resources (Parija et al., 2001) and taking advantage of water-based adhesives (Ahmadi-Dehnoei and Ghasemirad, 2021), which have gained a considerable deal of attention in a majority of industries, such as wood (Mabrouk et al., 2020), footwear (Orgilés-Calpena et al., 2019), and food packaging industries (Canellas et al., 2017). After drying, water-based synthetic adhesives could not be redispersed in water anymore; therefore, several techniques have been proposed for adhesive debonding with the aim of easy recycling and repairing of bonded

structures (Banea, 2019; Liu et al., 2020). These strategies include usage of solvent, heat, light or incorporation of dynamic chemical bonds or fillers in the adhesive (Banea, 2019). Given that in the past few decades, about 65 % of adhesives have been used in the wood industry (Liu et al., 2020), the development of dismantlable adhesives is very important in this field. In this regard, efforts such as usage of thermally expandable nanoparticles or dynamic disulfide bonds in the adhesive formulations were made to develop dismantlable adhesives for interior furniture and wood boards (Ishikawa et al., 2005; Liu et al., 2020).

Polysaccharides, lignin, tannin, and proteins are examples of bio-based polymers with potential application in wood adhesives (Norström et al., 2014). Polysaccharides are the most abundant category of bio-based polymers made of monosaccharides linked by glycosidic bonds. The high molecular weight of polysaccharides results in an acceptable cohesive strength of their adhesives. On the other hand, high contents of hydroxyl polar groups afford strong wood adhesion by creating hydrogen interactions. However, polysaccharide adhesives suffer from low water resistance due to their hydrophilic nature provided by high hydroxyl contents (Araújo et al., 2016). Therefore, in order to achieve suitable properties, their modification through blending or grafting with synthetic polymers, oxidation, esterification, crosslinking, or addition of inorganic

* Corresponding author.

E-mail address: ghasemirad@modares.ac.ir (S. Ghasemirad).

<https://doi.org/10.1016/j.indcrop.2021.114083>

Received 17 May 2021; Received in revised form 16 September 2021; Accepted 22 September 2021

Available online 14 October 2021

0926-6690/© 2021 Elsevier B.V. All rights reserved.